1

Toward a Smarter Future: The Role of AI in Transforming Automation Systems

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Abstract

The Intelligent automation systems (IASs) is the the powerful combination of artificial intelligence (AI) and automation system. This work examines the central function of AI as the driving force at the back of IAS to explore its actual global function, benefits, drawbacks, and moral concerns. Through research findings and realistic examples, we also investigated how AI amplifies the skills of IAS, improving efficiency, optimizing work-

flows, and enhancing decision-making across numerous industries. This study showcases the flexibility of IAS by highlighting its capacity programs in sectors including manufacturing, finance, and healthcare. The advantages of IAS including expanded productivity, cost savings, and reduced errors, additionally with capacity challenges, process displacement and protection dangers are acknowledged. Furthermore, we critically assess the ethical implications of IAS, which include data privacy, algorithmic bias, and societal effects. Emphasizing the importance of responsible improvement and governance frameworks for AI, we provide a comprehensive analysis of IAS, supplying treasured insights into the synergy between AI, automation, and the ethical considerations that shape their implementation and destiny development.

***Keywords*:** Automation, intelligent automation system, artificial intelligence, ethical issues, ethical considerations, applications

1.1 Introduction

While the concept of intelligent automation systems (IASs) might also seem novel, its origins may be traced again to two distinct narratives: artificial intelligence (AI) and automation. The roots of this fusion date back to the 1950s, when AI studies began, and the 20th century witnessed massive advancements in automation, together with meeting traces. In the 1990s, AI started to locate realistic applications, and with the aid of the 2000s, with the arrival of extra effective computing talents, AI and automation began to converge, in the long run, main to the emergence of IAS in the 2010s [1]. AI endows automation with the capacity to make selections, adapt, and analyze complex facts, thereby revolutionizing the way machines perform and shaping the future of shrewd automation. IAS represents a great evolution in automation, transcending the constraints of machines merely following predefined instructions. Instead, IAS leverages AI to create a greater wise and dynamic form of automation [2], wherein machines can analyze, purpose, and make decisions based on statistics and context.

At the heart of IASs lies AI, which endows machines with human-like skills which include gaining knowledge of, problem-fixing, and decision-

making. The AI thing inside IAS encompasses numerous natural language processing (NLP), such as machine learning algorithms that could pick out styles and continuously beautify their overall performance. Another facet of AI is NLP, which allows machines to apprehend and interpret human conversation, as illustrated in Figure 1.1. Additionally, laptop vision performs a crucial function by equipping machines with the functionality to research and interpret visible facts, thereby augmenting their potential to technique and make feel of visible records.

Within IASs, AI is not the only driving force; automation technology additionally plays an equally critical function. This comprehensive category contains a diverse array of equipment and techniques designed to automate responsibilities traditionally accomplished by human beings [2]. IASs embody a transformative method of automation, transcending traditional robot systems that function based on pre-programmed commands. By leveraging the power of AI, IAS set up advanced and adaptive automation surroundings, as illustrated in Figure 1.2. In the subsequent section, we will delve into the essential components that form the spine of IAS functionality, allowing their superior skills.

Artificial Intelligence (AI): IASs can be metaphorically compared to intelligent players, with AI acting as the intellectual engine that powers them enabling IAS in IAS to mimic human skills such as learning, problem-solving various solutions, and decision-making. The AI side manifests itself in

formance. NLP enables IAS to hear that human interactions are well understood and responded to. In addition, computer vision enables the IAS to “map” its surroundings by interpreting and analyzing visual information.

Automation Technologies: A team of automation technologies to support the AI component plays a key role in implementing AI decisions in IAS. Robotic process automation plays a key role in automating routine tasks with data binding and software communication, including streamlining processes and increasing productivity. It also focuses on flexibility, among the tasks to improve the overall efficiency of the IAS to ensure smooth ex-

ecution. These automation technology and AI components are powerful and synergistic alliances that work together to sustain the intelligence and transformational capabilities characteristic of IAS.

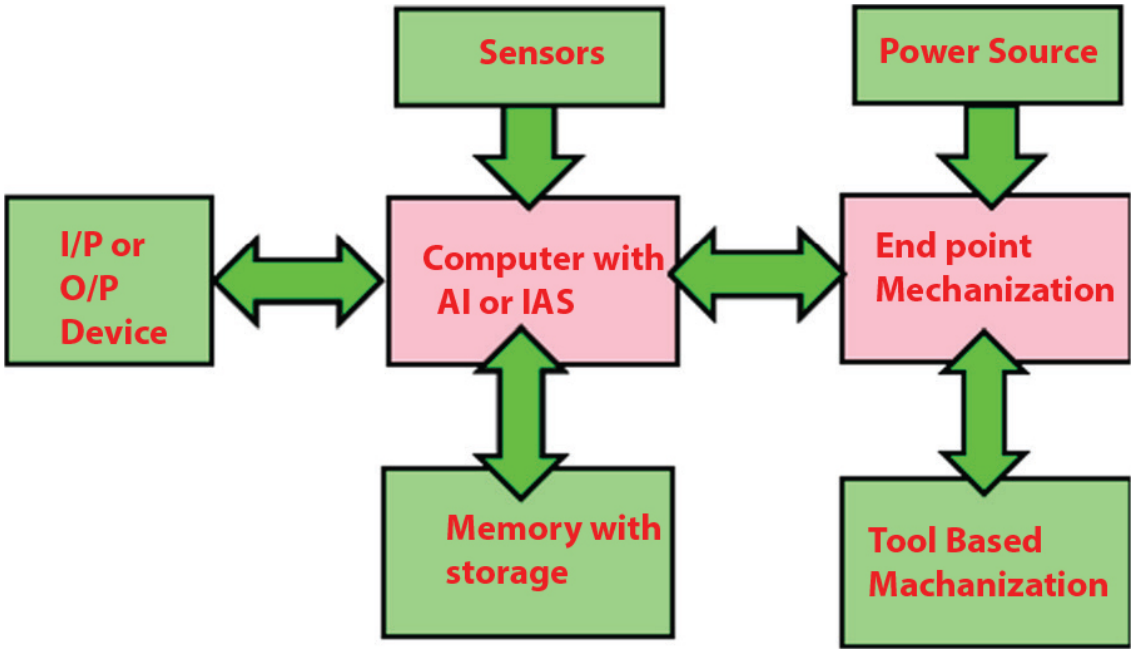


Figure 1.1 Block diagram of IAS.

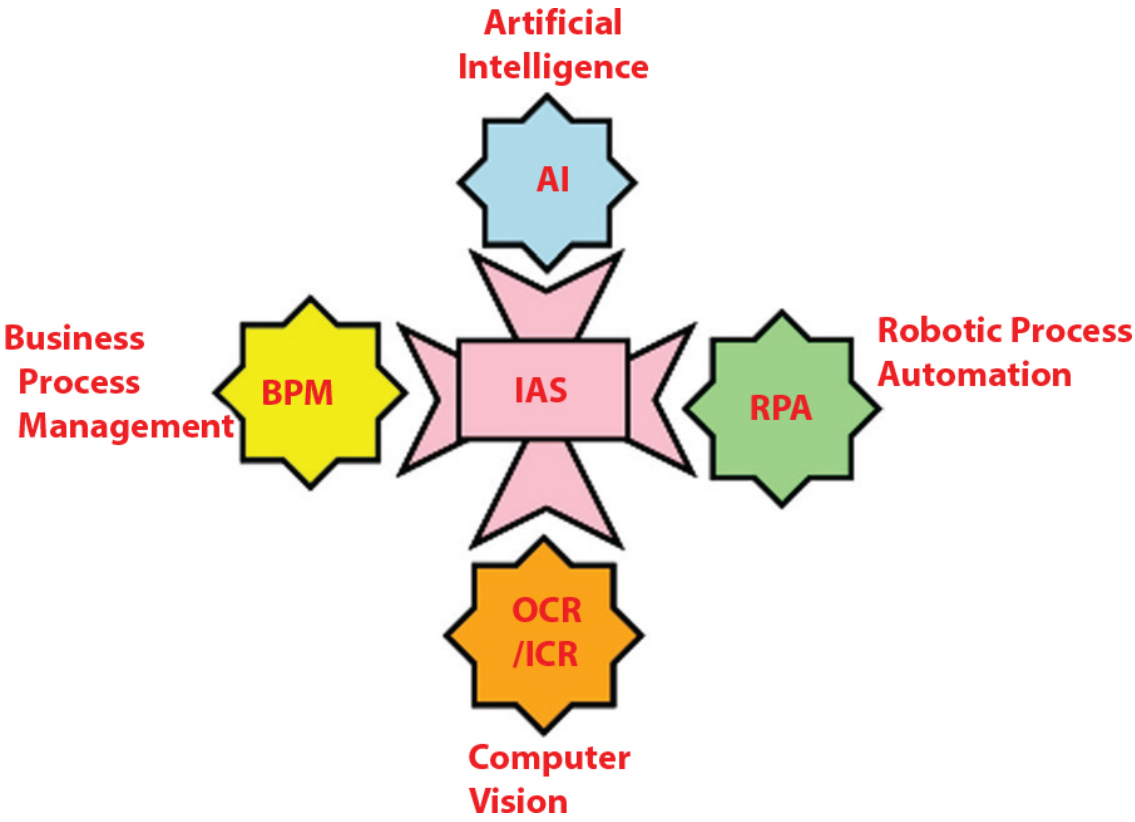


Figure 1.2 Key component of IAS.

To survey the landscape of IAS, with special emphasis on analyzing the key role and impact of AI in these systems. The chapter examines how AI contributes to the intelligence of IAS, increasing their capabilities in areas

such as training, decision-making, and adaptation. In addition, it addresses potential challenges to IAS, such as job outages and security risks, to provide a comprehensive understanding. Additionally, the chapter deeply examines the ethical concerns surrounding AI-powered automation, such as data privacy, algorithmic bias, and the broader societal implications of spreading AI adoption. By addressing these issues and emphasizing the importance of developmental responsibility, the chapter seeks to contribute to a future in which IAS values all stakeholders in direct improvements and ethical, automated technology while mitigating potential risks and negative impacts.

1.2 The Power of AI in IAS

The decision making power in IAS system occurs from the AI that elevate it from the basic automation to the potential of examine, adapting, and decision making [3]. The AI issue inside IAS demonstrates itself in numerous paperwork, along with gadget learning algorithms that become aware of patterns and constantly improve, permitting IAS to expect results and optimize methods. Moreover, NLP empowers IAS to comprehend human conversation, even as computer vision permits them to their surroundings, facilitating duties like item popularity. In essence, AI within IAS serves as the catalyst that propels traditional automation to extraordinary stages of energy and dynamism, transforming it into a wise and adaptive gadget [4]. Figure 1.3 shows the power of the IAS block diagram.

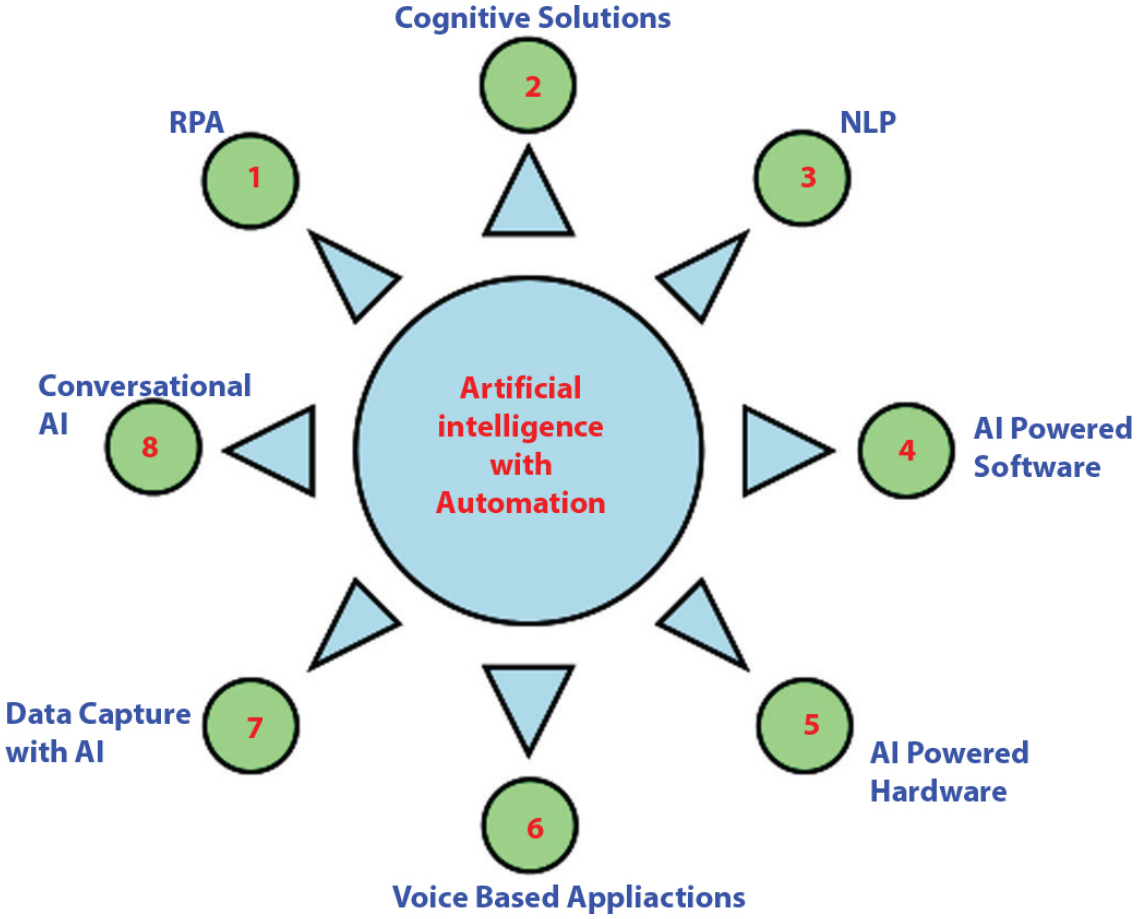


Figure 1.3 Power of the IAS block diagram.

Within IASs, AI demonstrates itself in various forms, including device learning algorithms that can expect disasters in manufacturing facility equipment or decorate the overall performance of Chabot's in expertise consumer queries [5]. Deep learning similarly extends AI's capabilities, permitting it to address complicated duties along with facial reputation for safety functions and analyzing enormous portions of facts to customize advertising and marketing techniques. NLP empowers machines to interact in conversational interactions, from virtual assistants answering questions to document processing systems comprehending textual facts [6]. Computer imagination and prescience equip IAS with the capacity to visually understand and analyze their surroundings, from visitor tracking systems to self-driving cars navigating the arena around them. These examples represent simply a fragment of AI's capability applications inside IAS, and as AI keeps conforming, it promises to release even extra superior and shrewd automation abilities [7].

1.3 Transforming Automation: A Multifaceted Impact

AI serves as the primary cognitive engine that powers IASs, endowing them with the ability to research, adapt, and make balanced choices. Machine getting to know algorithms inside IAS allows those structures to examine extensive datasets, become aware of patterns, expect future activities, and constantly decorate their overall performance. For example shown in Figure 1.4 of the working module of IAS, manufacturing unit machines ready with AI can analyze sensor statistics to expect ability gadget failures, thereby preventing highly priced downtimes [8].
Furthermore, AI bestows IAS with choice-making competencies, wherein NLP allows them to comprehend human conversation and make informed choices primarily based on user input, as illustrated in Figure 1.4.
In principle, AI transforms IAS from fundamental robotic systems into intelligent entities capable of gaining knowledge of, adapting, and making selections, basically reshaping the landscape of automation across various industries.

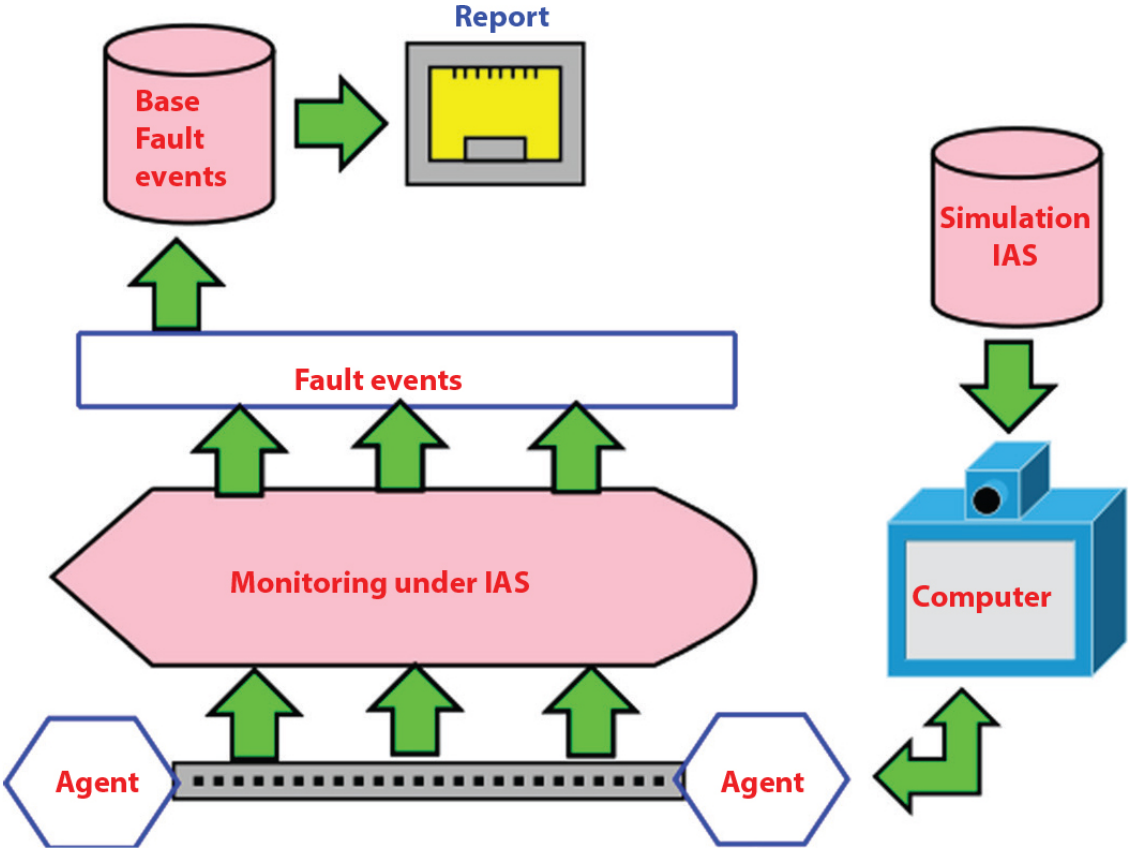


Figure 1.4 Working module of IAS.

AI-driven IASs act as efficiency supporters, automating repetitive tasks and releasing human assets for extra strategic duties. Within IAS, machines gain knowledge of algorithms characteristic as information analysts, reckoning out blocks in workflows and permitting the systems to goal inefficiencies, get rid of redundant steps, and optimize processes for more desirable speed and accuracy. For example, AI can streamline bill processing by means of extracting crucial facts and hastily routing it for approval. Similarly, AI-powered chatbots can deal with habitual purchaser inquiries, permitting human marketers to concentrate on more complicated problems. In essence, IASs function as diligent and intelligent assistants, enhancing efficiency and streamlining processes throughout diverse industries, saving sizable time and resources in the long run [9, 10]. These structures act as tireless people, relentlessly identifying regions for improvement and implementing automatic solutions, allowing groups to operate with greater agility and productiveness.

The impact of IAS is thoughtful, with studies predicting that AI-powered automation may want to potentially rise worldwide productiveness by a super $1.5 trillion by way of 2030. Real-international examples across diverse industries validate these blessings, as visible within the transforma-

tive outcomes of IAS in production [11]. The integration of device mastering and laptop vision in IAS has extensively better excellent management, enabling superhuman accuracy in product inspections and lowering illness costs. Similarly, the healthcare zone is presenting a process of change, with AI chatbots helping patients and assuaging the workload of clinical professionals. Furthermore, customer support has experienced a shift, with digital assistants powered by NLP managing habitual duties, resolving problems, and personalizing interactions, in the long run, main to accelerated purchaser pride [12]. These examples display the long-way-accomplishing ability of IAS, permitting organizations to perform with greater performance, accuracy, and patron-centric strategies throughout diverse sectors.

1.4 Benefits and Impact of IAS

IASs are driving transformative changes across various sectors by enhancing efficiency and boosting productivity. By automating repetitive tasks, such as processing loan applications in the banking industry, IASs empower human workers to shift their focus toward higher-value responsibilities, such as conducting in-depth financial analyses and delivering exceptional customer service experiences. This strategic allocation of human resources, facilitated by IAS, enables organizations to optimize their operations and unlock greater value from their workforce.

It may be regarded as notably gifted assistants, significantly boosting productivity, precision, and value financial savings throughout diverse industries, as illustrated in Figure 1.5. The integration of AI within IAS allows the evaluation of statistics to discover unseen charms, consisting of predicting fitness dangers in patients, thereby facilitating more knowledgeable decision-making, ranging from preventive healthcare to electricity-saving projects in homes based totally on utilization records. The impact of IAS extends past factories (predictive protection) and banks (fraud detection) to redefine business operations in retail (personalized studies) and other sectors. Furthermore, IAS can make contributions to enhancing public services (streamlined paperwork processing), social applications (focused help), and public safety (predictive crime analytics) even fostering citizen engagement. However, it is far vital to address chal-

lenges consisting of job displacement and cybersecurity issues to make sure a accountable and fantastic future driven via the adoption of IAS. By harnessing the strength of AI and automation in an accountable way, IASs have the ability to revolutionize numerous elements of society and power giant improvements in efficiency, accuracy, and selection-making techniques.

It generates cost financial savings through two primary mechanisms: automating tasks that do not require human intervention and leveraging AI to discover untapped possibilities for financial savings, such as studying constructing energy usage to pinpoint regions requiring improvement [13]. This procedure liberates monetary sources, enabling vital investments in regions that include social packages or infrastructure improvements. IAS minimizes human mistakes, as AI systems can process extensive quantities of statistics with fantastic precision, thereby reducing inaccuracies in domain names like finance, public data, and felony files. Consequently, IAS fosters a more transparent, accountable, and straightforward government, complemented by elevated productivity, value financial savings, and a progressed citizen career experience. IAS represents a transformative force for governments aiming to carry out with more intelligence and performance, optimizing their resources and delivering higher fees to citizens.

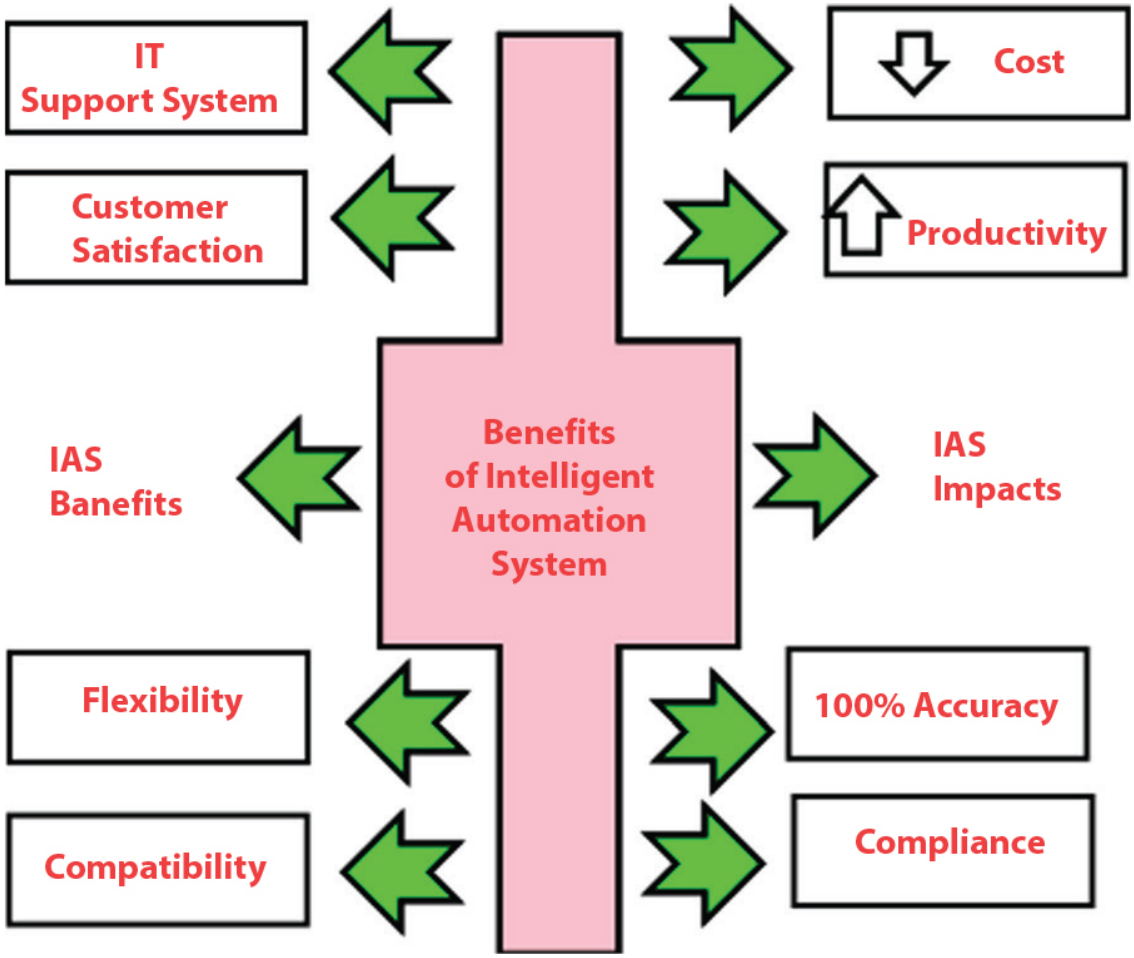


Figure 1.5 Benefits and impact of IAS.

1.5 The Spectrum of Applications: From Manufacturing to Beyond

IASs are transforming several industries via leveraging AI and statistics assessment. Let us explore how IAS is reshaping unique domain names:

1.5.1 Predictive Maintenance

IAS is revolutionizing factory operations by utilizing AI to research sensor records from machines. This proactive method enables IAS to predict capacity device screw-ups earlier than they stand up, ensuring uninterrupted operations and stopping highly priced downtimes. For instance, within the aviation industry, IAS can examine engine information to anticipate preservation necessities, thereby improving flight safety and punctuality.

1.5.2 Finance

Financial institutions are embracing IAS prepared with AI competencies to meticulously examine considerable quantities of transaction information, allowing real-time detection of suspicious activities to defend clients and prevent economic losses. For example, banks can use IAS to perceive uncommon credit score card spending patterns that may indicate fraudulent sports.

1.5.3 Retail

Retailers are utilizing IAS to gain insights into purchaser alternatives. By reading shopping for conduct, IAS can endorse merchandise that customers are probably interested in, advise targeted promotions, or even modify expenses dynamically. Imagine online stores suggesting garb based totally on your preceding purchases.

1.5.4 Healthcare

IAS with AI is an alternative to the healthcare enterprise by way of helping clinical professionals. It can analyze scientific photographs for quicker and better accurate diagnoses and even use affected person records to personalize treatment and expect fitness dangers. Imagine AI improving the detection of breast cancers in mammograms, a promising vicinity of studies.

1.5.5 Logistics and Supply Chain

IAS helps transportation businesses come to be development controllers. By analyzing visitors and weather facts in actual time, IAS can optimize delivery routes, saving gas and slashing shipping instances. Imagine logistics organizations using IAS to discover the fastest routes for trucks, getting applications to customers faster than ever [13].

1.5.6 Customer Service

IASs are customer service superheroes! Chatbots powered by means of IAS can solve questions and resolve basic issues around the clock, liberating human dealers from tougher problems. Virtual assistants may even customize interactions, making customer support more handy and efficient [1, 2]. This is only a glimpse of what IAS can do, and as AI receives even smarter, we are able to anticipate even more first-rate packages throughout industries, changing how we work and riding development inside destiny [13].

1.6 Challenges and Considerations

Releasing the efficiency of IAS is awesome, but we need to consider some challenges before diving in [13]. Here's a look at key points we need remember.

1.6.1 Technical Hurdles

Getting IAS to paint with all of your existing systems can be complex. For example, a central authority organization looking to apply AI for recruitment may face challenges if their old worker data are not compatible. To keep away from this, facts need to be standardized so that everthing communicates smoothly [14].

1.6.2 Cybersecurity Threats

Since IAS uses tons of records, it is like a cyber treasure chest—robust cybersecurity and moral records practices are key to retaining the whole lot secure [13].

1.6.3 Skill Gap

Building and running IAS is not always like using a toaster—it calls for specialized capabilities in records, AI, and cybersecurity. Investing in schooling and acquiring these tech capabilities is vital to getting the maximum out of IAS [13].

1.6.4 Ethical Concerns

AI is not best—biases in education data can cause unfair decisions, like who gets a mortgage or how the law is enforced. To keep away from this, we need to cautiously pick records and continuously monitor AI structures to maintain things fair and impartial [15].

1.6.5 Transparency and Explainability

Different from humans, AI structures may not constantly offer clear explanations for their selections which can make it tough to agree with their conclusions, particularly in important domain names like law enforcement or medication. However, explainable AI strategies can shed light on the reasoning behind AI's decisions, enabling us to recognize the reason and foster agreement within the method [16].

1.6.6 Social and Regulatory Considerations

For IASs to be without a doubt a hit, it is far crucial to build acceptance as true among customers. Transparent communication about the abilities and obstacles of IAS, coupled with proactively addressing moral concerns, is vital. Public focus campaigns and academic projects can play a sizeable function in fostering this agreement [17].

1.6.7 Regulation and Governance

As AI advances swiftly, so it is far more important to establish properly described guidelines. Governments should aim to strike stability between encouraging AI innovation and mitigating capability risks, such as activity displacement or privacy concerns.

1.6.8 Job Displacement

While IASs allow automation to create new opportunities, additionally, they introduce challenges for the personnel. Jobs concerning ordinary duties, such as fact entry, meeting traces, or sure customer service roles, can be prone to automation. For example, AI chatbots could manage simple

purchaser inquiries, probably replacing some customer support representatives. This automation can lead to unemployment, financial hassle for affected employees, and emotional outcomes of process loss.

1.6.9 Workforce Impact

Automation, pushed by IAS, has a blended effect on employees, appearing as a double-edged sword. While ordinary jobs may be changed via AI chatbots or assembly line robots, new possibilities will emerge. The key component lies in adaptability. Governments and academic institutions ought to spend money on training packages for facts science, digital literacy, and crucial wondering to facilitate people's transition. These acquired competencies might be essential for the roles generated through AI, consisting of engineers responsible for designing and retaining these superior structures.

1.6.10 Income Inequality

Automation can also exacerbate profit inequality, as the new jobs generated by using AI may require superior education and abilities, inadvertently leaving those without access to schooling possibilities even similarly at the back of [18].

1.7 Strategies for Mitigating Negative Impacts

To put together workers for the evolving automation landscape, governments and organizations must allocate resources toward schooling projects, consisting of apprenticeships, vocational training, and ongoing expert improvement packages [15, 19–21].

1. Universal Basic Income (UBI): As a response to the body of workers' transformation due to automation, some specialists advise implementing a universal basic income (UBI) as a safety internet for workers displaced with the aid of those modifications. UBI might offer a guaranteed minimal income for all residents, regardless of their employment repute.

2. Labor Market Policies: Governments can mitigate the impact of automation by supplying unemployment benefits, process retraining programs, and rules that encourage the introduction of recent positions in sectors experiencing growth because of automation. Destiny involves a collaborative attempt between people and wise machines. By acknowledging ability activity losses as a result of automation and enforcing measures to help employees in transition, which include retraining applications, we can make certain that the benefits of this technology are equitably shared, leading to a future where all stakeholders prosper in the long run.
3. Security Vulnerabilities: Security vulnerabilities can be likened to weaknesses in a gadget's defenses, offering capability access points for malicious actors to steal records, cause disruptions, or benefit from undetected entry. These vulnerabilities can exist in hardware, software, firmware, or maybe in the interactions among users and the system. Let us categorize and take a look at numerous safety vulnerabilities that can be present in structures:

   Software Bugs: Imperfections in code, which include typographical errors or extra large design flaws, can create security vulnerabilities that malicious actors may also take advantage of.
   System Misconfigurations: Failing to secure systems with robust passwords, closed ports, or enabled safety capabilities is similar to leaving doorways unlocked, providing malicious actors with smooth entry factors.
   Zero-Day Attacks: Zero-day exploits can be likened to silent attacks, targeting unidentified vulnerabilities for which no shielding measures, which include patches, had been superior, thereby increasing their potential for damage.
   Social Engineering: Social engineering assaults can be likened to digital deceptions, aiming to manipulate individuals into disclosing private facts or attractive with malicious links, often via phishing emails or junk mail [22].

1.8 Ethical Considerations of IAS

While the advantages of IASs are interesting, it's far important to cautiously remember sure moral issues earlier than completely embracing

this generation. Let's observe the factors that authorization careful attention:

1.8.1 Bias and Fairness

AI can unintentionally include biases if its training facts isn't carefully curated, main to inequitable decisions, which consist of those associated with loan approvals or hiring. To mitigate this, it's far important to lease a severa array of records, constantly show AI structures for bias, and intentionally are searching for out heterogeneous datasets to sell sincere training.

1.8.2 Transparency and Explainability

AI's preference-making method can be a good deal less transparent than human choice-making, making it tough to don't forget its picks. However, explainable AI techniques act as a spotlight, illuminating the common sense behind AI's picks and fostering public be given agree with.

1.8.3 Privacy Concerns

Despite the remarkable advantages provided by way of IAS, their reliance on facts increases privateness concerns. Implementing stringent information security features and ethical practices is crucial to ensuring protection. This includes empowering people with manipulate over their records and supplying them the choice to choose out of records collection. By prioritizing fairness, transparency, and privateness, we can ensure that IAS is used for the more appropriate, paving the manner for a future in which technology uplifts anyone.

1.8.4 Mitigating Bias and Ensuring Fairness

Imagine an AI loan officer by accident discriminating based on race or gender because its schooling data turned into biased. Responsible improvement method the usage of a lot of records and constantly checking AI structures for fairness to save you such situations, ensuring everybody gets a fair shot.

1.8.5 Building Trust Thru Transparency

Some AI algorithms are like mysterious black packing containers, which can be frightening. To build consider, responsible improvement desires tools that explain how AI makes choices (explainable AI) [16]. This transparency facilitates people recognize and take delivery of AI's picks, heading off situations where hidden formulation improve issues approximately fairness and who is responsible [16].

1.8.6 Safeguarding Privacy in the Age of Big Data

Since IAS relies on mountains of information, statistics privacy is a pinnacle problem. Strong rules (governance frameworks) are critical to hold statistics secure. This method difficult cybersecurity, clean data use policies, and placing people in control of their very own facts. By prioritizing privacy, we will build believe and ensure personal facts is used responsibly.

1.8.7 Addressing Job Displacement and Reskilling Needs

While IAS automation might result in activity losses in some regions, responsible development can help us navigate this. Clear guidelines (governance frameworks) ought to prioritize retraining and upskilling applications for workers [21]. This allows human beings whose jobs are affected adapt and transition smoothly to new opportunities in the evolving process market.

1.8.8 Shaping the Future with Ethical Considerations

Unleashing IAS is exciting, but we need guardrails! Strong rules (governance frameworks) are key to considering the impact of AI on society, like job losses, AI in war, or even social manipulation [12–15]. By prioritizing ethics, we can ensure IAS is used for good. This isn't optional - responsible development and clear rules are essential to making IAS a positive force. Think fairness, transparency, data privacy, and ethics – with these in mind, IAS can be a tool for a better future, where technology uplifts everyone [22].

1.9 Discussion

AI is breathing intelligence into automation, turning rigid robots into adaptable learners. This promises a future brimming with efficiency, from smarter factories to personalized medicine [13, 23–27]. However, we need to tread carefully, considering ethics and job displacement. By focusing on responsible development and reskilling workers, we can unlock a future where humans and AI collaborate for the greater good as Figure 1.6 shows the revolutionizing automation with AI and Figure 1.7 depicts the analytical progress of IAS.

Get ready for a world where smart machines handle the boring stuff, powered by a dream team of AI and other cutting-edge tech. This intelligent automation revolution will free us to unleash our creativity and innovation [13, 23–27].

Intelligent automation isn't just about robots doing chores - it's approximately AI smarts taking on complicated tasks. Imagine software program robots that analyze, make decisions, and even chat with customers. Intelligent automation structures free us from habitual obligations, allowing us to recognize more strategic and creative endeavors. The advantages expand beyond office settings. In the healthcare domain, AI can facilitate advanced disease diagnosis and tailor treatments to male or woman sufferers. Even in our houses, clever automation can optimize energy consumption, decorate security features, and personalize dwelling areas – ushering in a clever revolution that seamlessly integrates the era into our day-by-day lives [28–32].

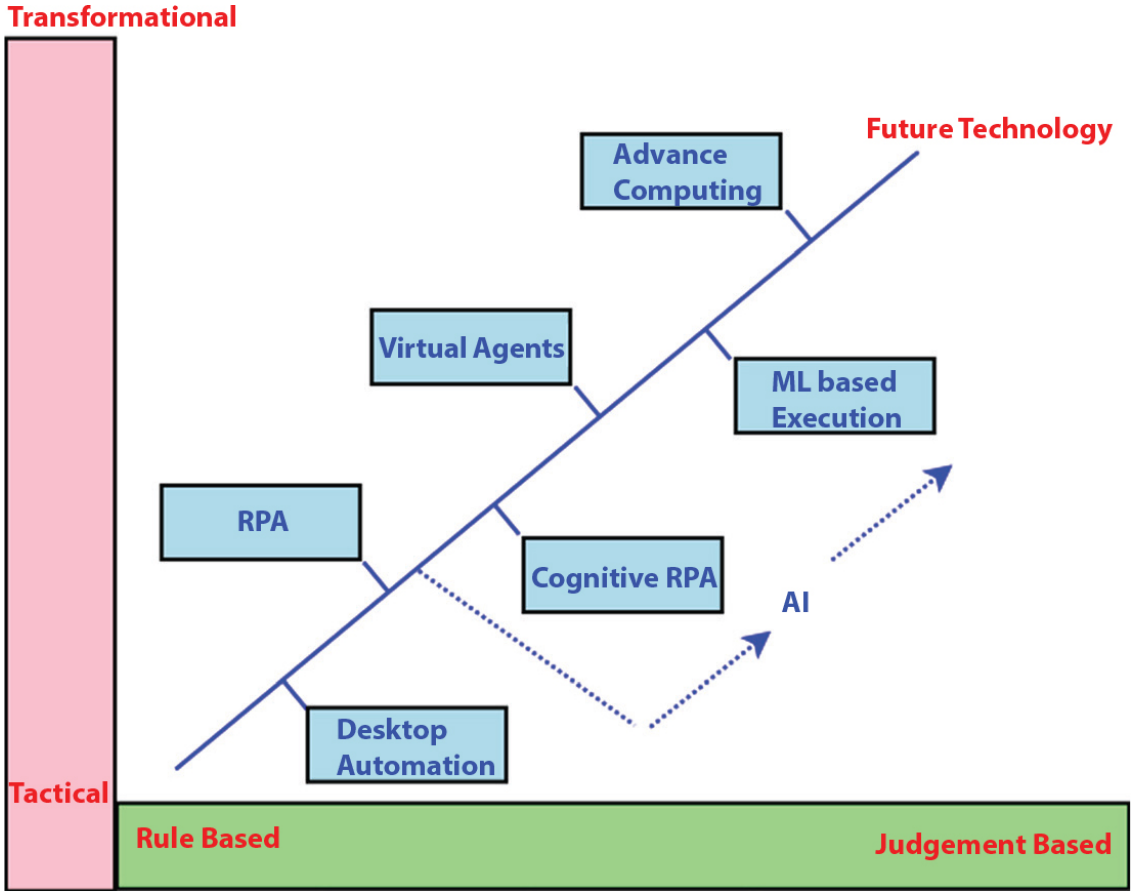


Figure 1.6 Revolutionizing automation with AI.

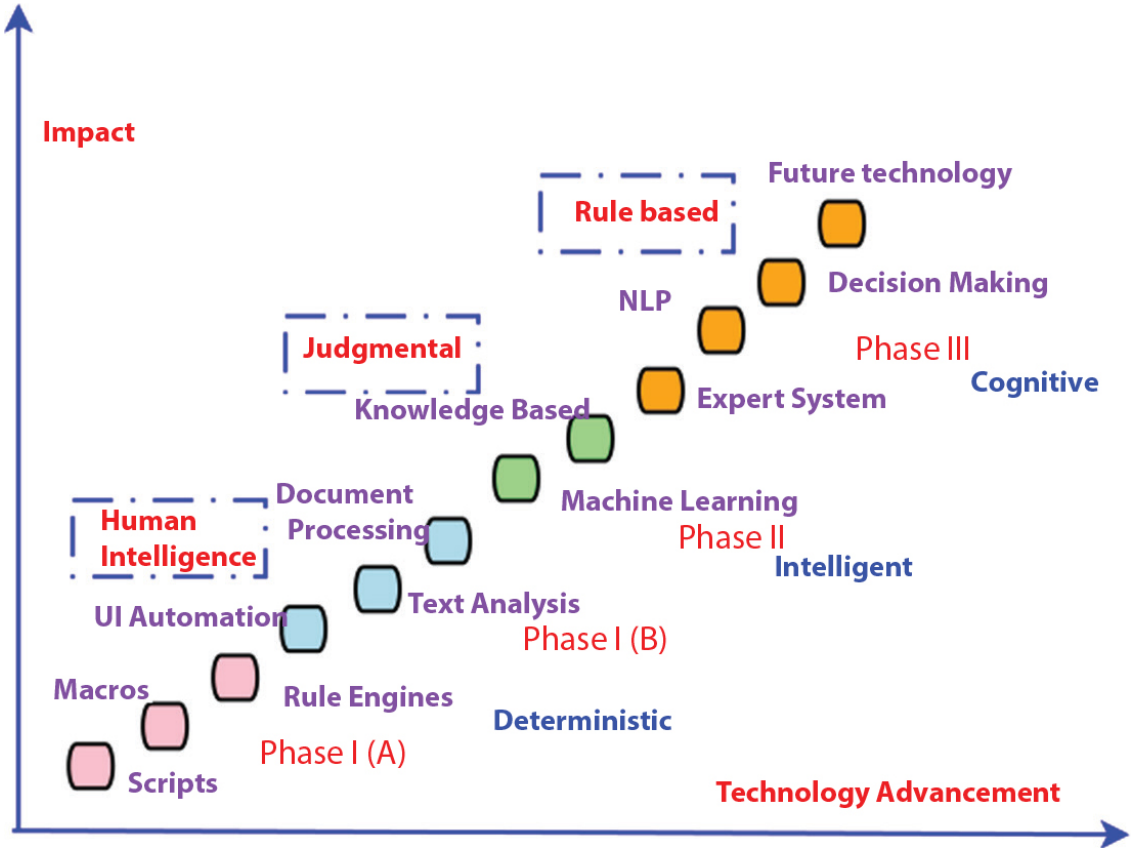


Figure 1.7 Analytical progress of IAS.

1.10 Conclusion

This study provides an insightful exploration of the charming world of IASs, wherein synthetic intelligence and automation converge. We delved into the transformative impact of AI on IAS, examining its applications throughout numerous industries. Through our research and real-global case research, we witnessed the profound effect of AI on IAS, allowing greater green strategies, improved productivity, and advanced selection-making competencies. Additionally, we examined the diverse capability of IAS to catalyze large changes for the duration of sectors consisting of production, finance, and healthcare. While IAS constitutes a transformative pressure in automation, imparting felicity like more potent overall performance, charge savings, and mistakes discount, it additionally introduces challenges, consisting of technique displacement and protection risks, which necessitate thoughtful control. In addition explored the moral elements of AI automation, addressing problems like facts privateness, bias, and societal outcomes. IAS are effective gear, and through emphasizing accountable improvement and organizing well-defined recommendations, AI in IAS can characteristic as a constructive pressure, making sure that the blessings of intelligent automation are shared equitably utilizing the use of all.

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Note

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